

Anomaly Detection in Material Images using Scan Statistics



Kaitlyn Anderson,⁺ Asanka Duwage,[‡] and Tung-Lung Wu [‡]

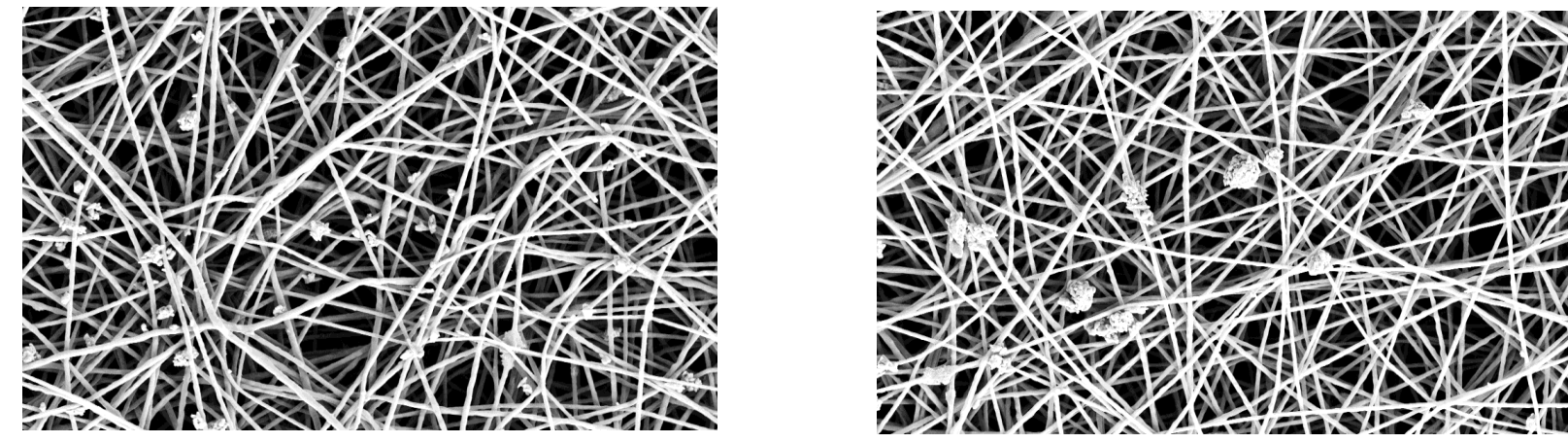
⁺ Department of Statistics, University of Michigan

[‡] Department of Mathematics and Statistics, Mississippi State University



Introduction

- Nanoproductions are an emerging sector with unique properties and diverse applications across industries¹
- Nanomaterial production requires a high degree of precision, making it challenging to ensure consistent quality at a large scale¹
- Minor defects can significantly alter nanomaterial properties and performance, making accurate defect detection crucial¹
- This project uses scan statistics to detect localized clusters in scanning electron microscope (SEM) images of nanofibrous materials²



- Scan statistics enable detection of localized anomalies and their locations, making them ideal for identifying spatial defects in SEM images that might be missed by global methods
- We implement and compare multiple scan-based methods to identify the most adaptive and robust detection approach

Scan Statistic

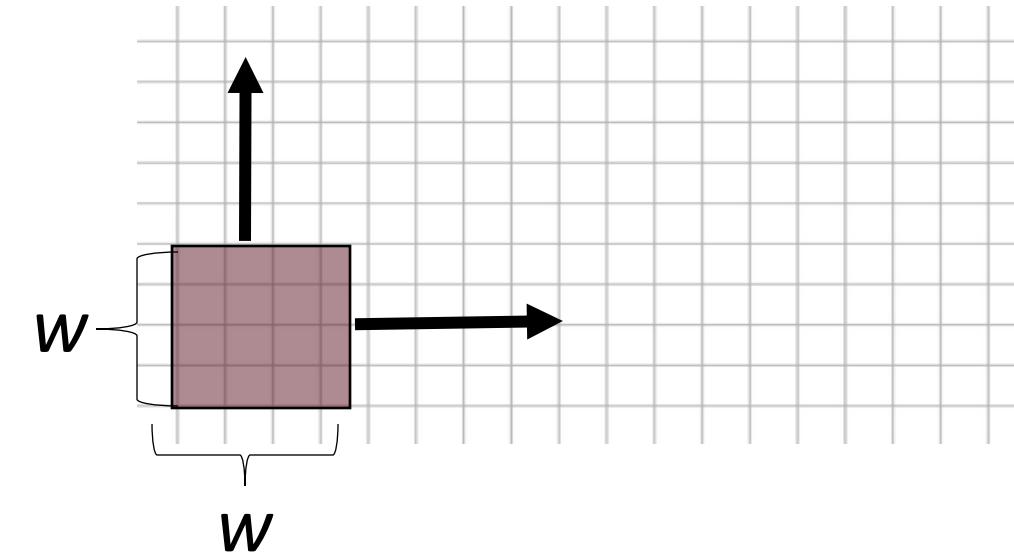
- Let w be the side length of a square scanning window. For each window position (i_1, i_2) , define:

$$Y_{i_1, i_2}(w) = \sum_{i=i_1}^{i_1+w-1} \sum_{j=i_2}^{i_2+w-1} X_{i,j}$$

as the total pixel intensity within a square region, where each element $X_{i,j}$ corresponds to the intensity at row i and column j in the matrix.

- The **scan statistic** for window size w in a $N_1 \times N_2$ image is defined as:

$$S_w(N_1, N_2) = \max_{\substack{1 \leq i_1 \leq N_1 - w + 1 \\ 1 \leq i_2 \leq N_2 - w + 1}} Y_{i_1, i_2}(w)$$



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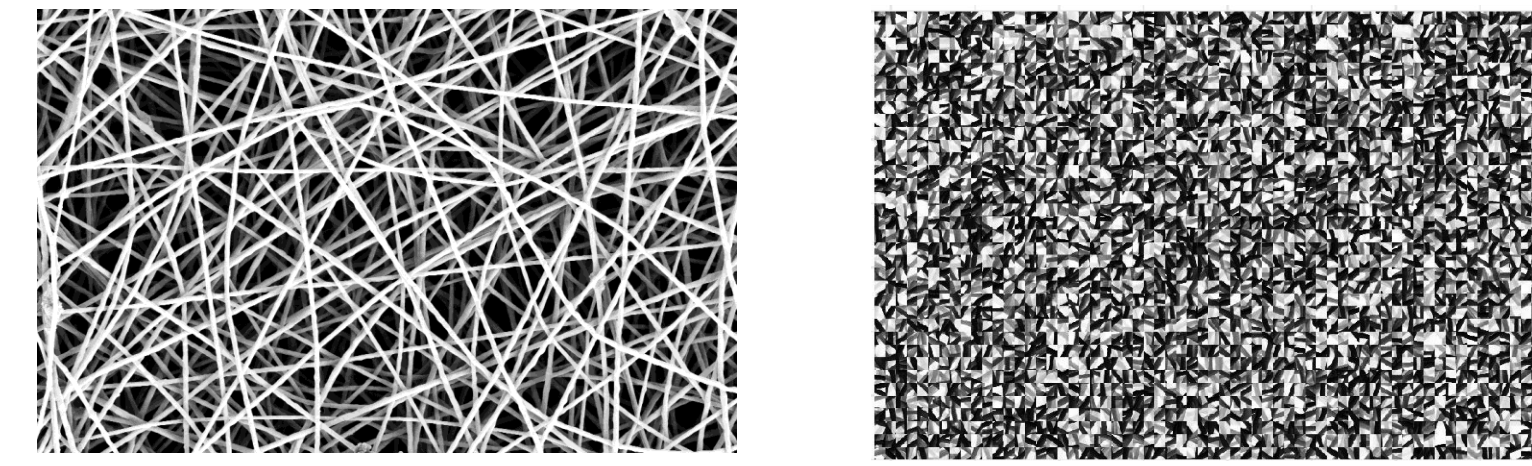
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HIGH PERFORMANCE
COMPUTING COLLABORATORY

References

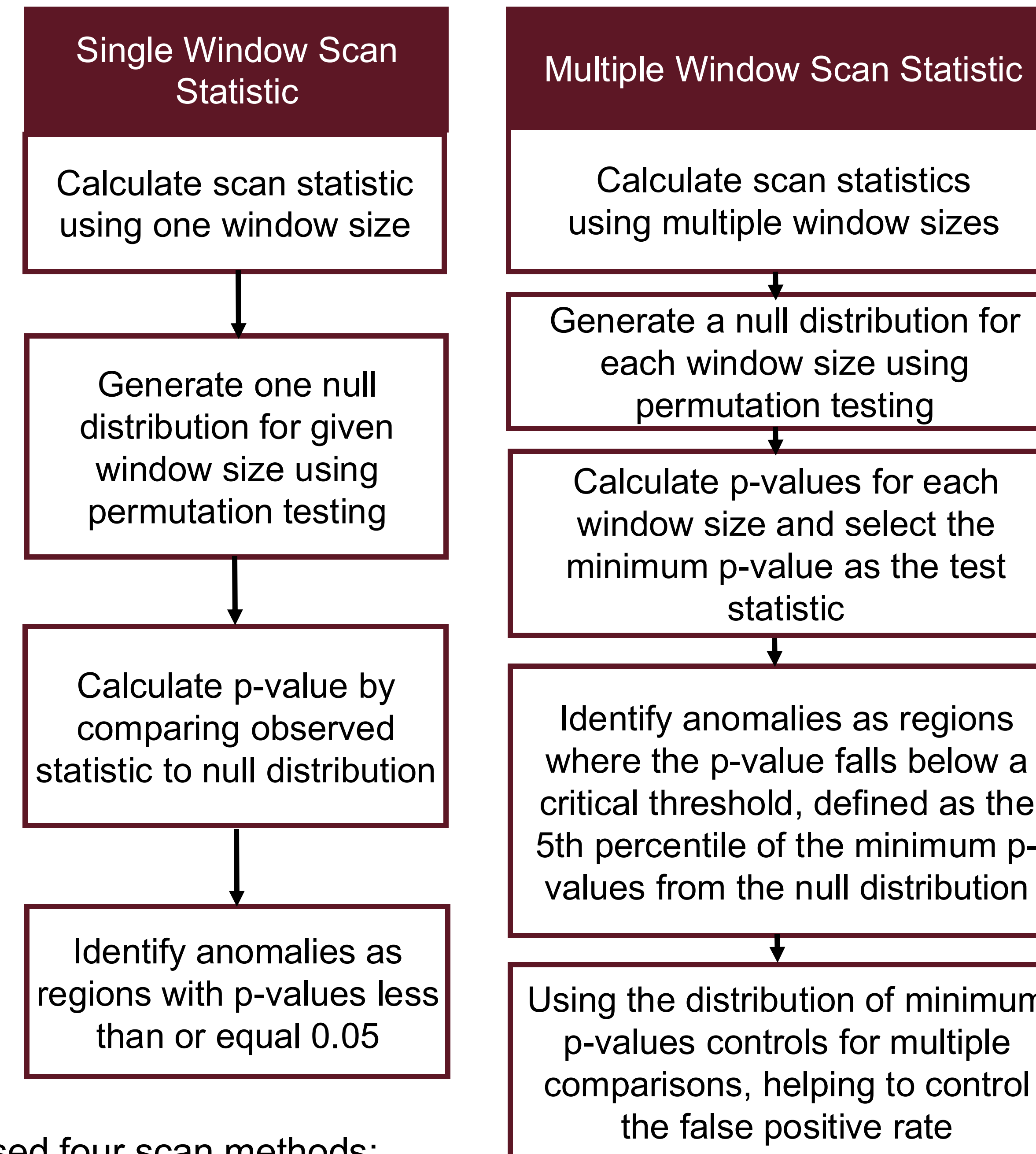
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Methods

- Dataset includes 40 anomalous and 5 normal SEM images, which are transformed into normalized grayscale matrices
- Perform a hypothesis test by comparing the scan statistic to a null distribution:
 H_0 : no spatial clustering
 H_A : significant spatial cluster present
- The null distribution is generated through permutation testing by dividing normal images into fixed-size regions (patches), permuting them, and computing the scan statistic for each permutation using a given window shape and size



- Both single and multiple window scan statistics were applied; the process works as follows: for a given anomalous image,



- Used four scan methods:
 - single-window square (50×50 window)
 - single-window circle (window with radius of 28)
 - multiple-window weighted square, where the scan statistic for each region is divided by the standard deviation within the window (20×20, 30×30, 40×40, 50×50, 60×60, 70×70, and 80×80 windows)
 - multiple-window unweighted square (same window sizes as above)

Results

- Power across permutation patch sizes using multiple-window square scan with the seven window sizes listed previously [1] and power across methods using 14x16 patch size [2]

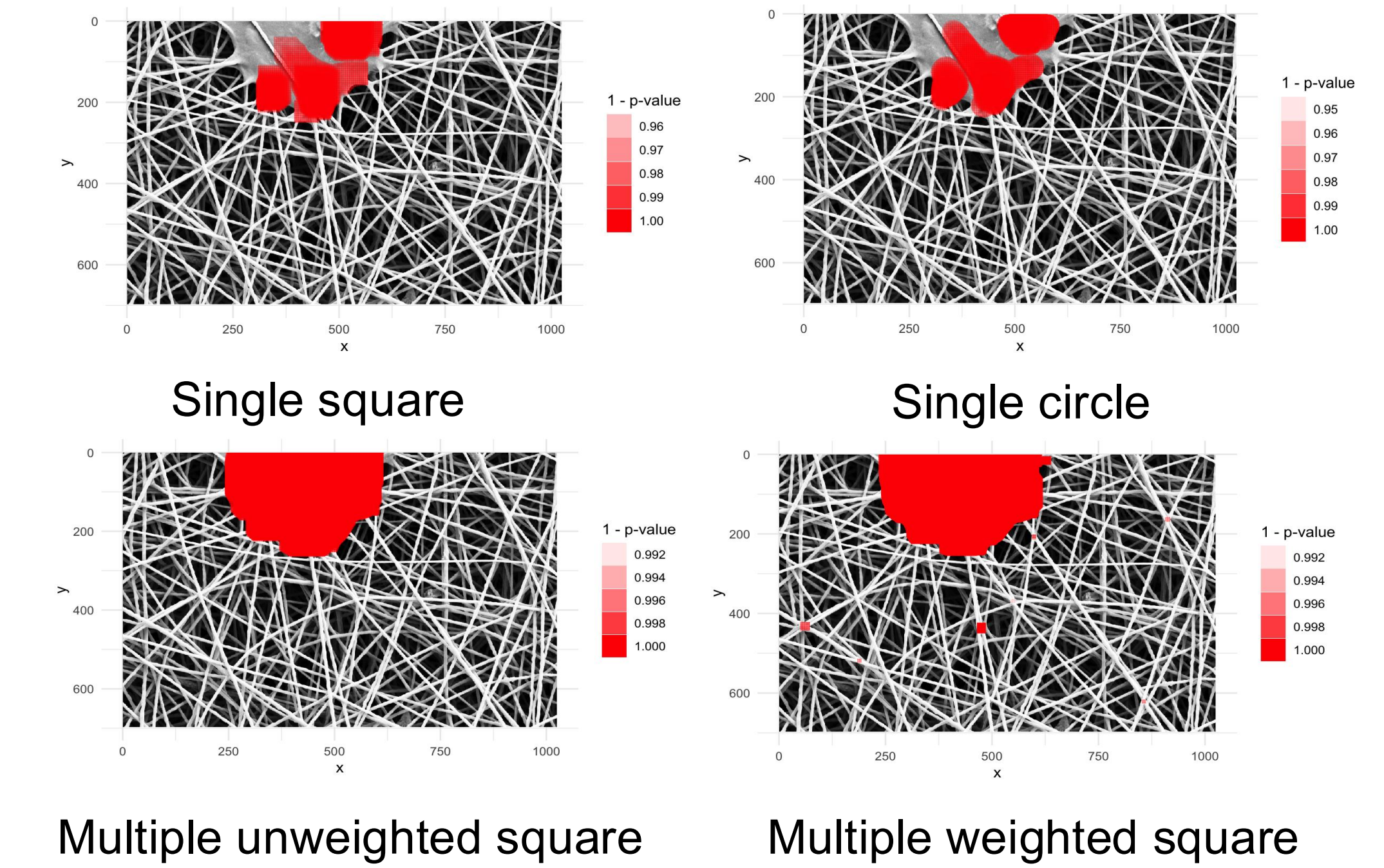
Patch Size	Power
14x16	32/40
28x32	19/40
70x64	18/40
140x128	23/40

Figure 1

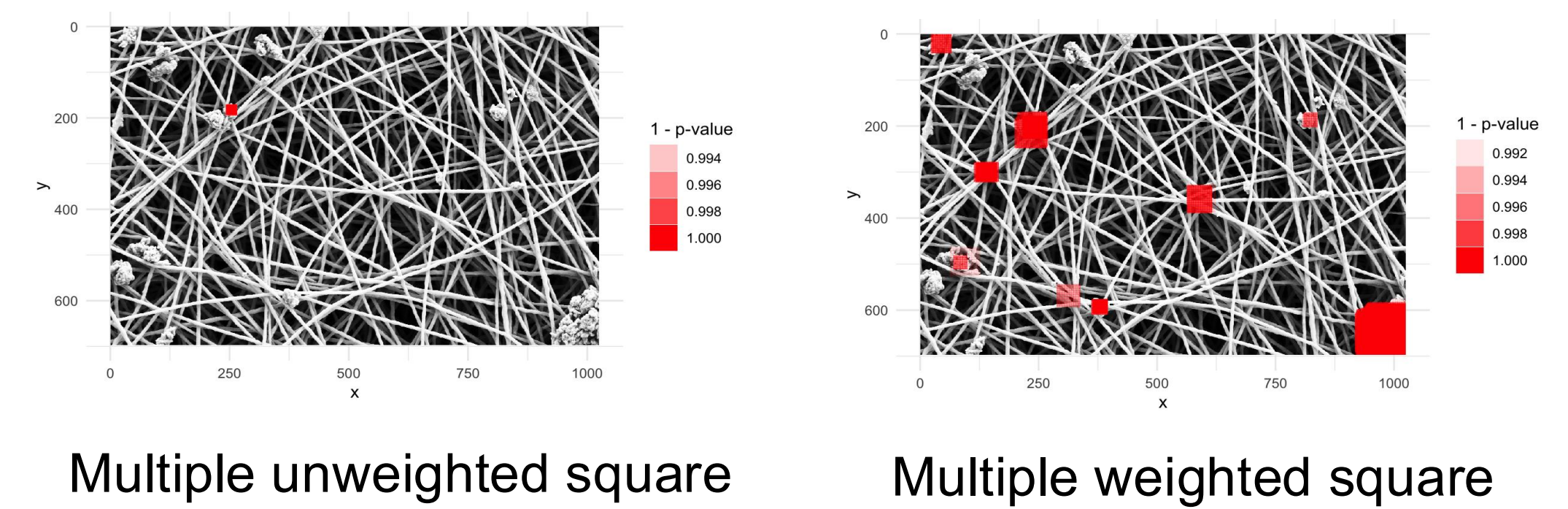
Method	Power
Single square	37/40
Single circle	33/40
Multiple unweighted square	32/40
Multiple weighted square	40/40

Figure 2

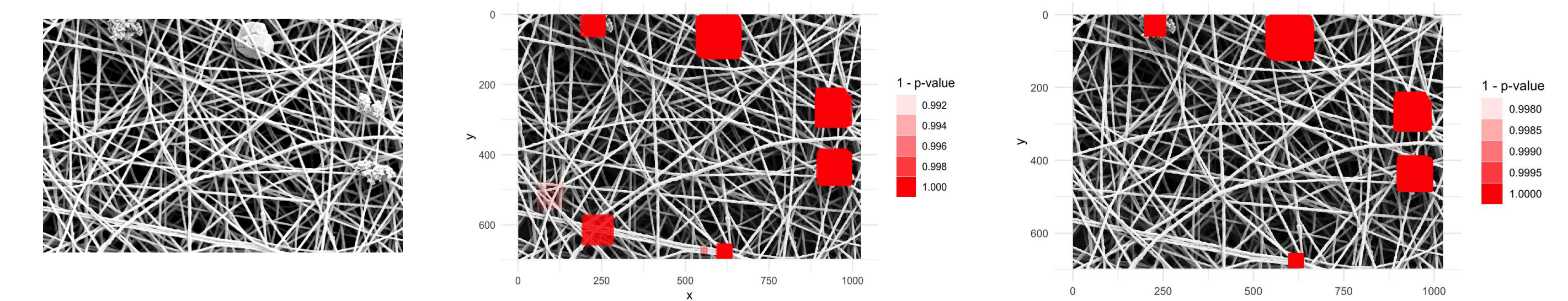
- Compared detected regions across four scan methods



- Assessed impact of weighted scan statistic



- Compared weighted scan statistic results using smaller significance level



Conclusions

- Multiple-window scans improve coverage by addressing the varying sizes and shapes of the anomalies
- Weighted scans enhance detection by using local variance to distinguish defects from equally bright normal regions and to address shadows on the defects in the images
- Lowering the significance level or displaying only the most significant regions improves detection accuracy by reducing false positives